Guide to Session 2

Ubiquitous, Yet Hard to See · BCH 3023 · binding in the NMR hall, a crystal in the X-ray room, and three structures compared

In Session 2 you record how two partner proteins bind ubiquitin, see how NOEs become a structure, then grow, freeze and shoot a crystal of ubiquitin, read its resolution and its electron density, and lay the crystal structure, the NMR structure and a prediction over one another. This guide takes you through every step: **what to do**, **what you should see**, and **why**. Open the lab page with Session 2 (the link under the title, or `?part=2`); your code is the same as in Session 1.

Before you start

- Read Part C of the prelab review (topics 11 to 15).
- Every step can be done in the 3D rooms or with the buttons beside them. The views under the room include the X-ray room's: `X-ray room` `Crystal bench` `Diffractometer` `Inside` `X-ray desk`.
- Allow about an hour and a half. The pictures here were taken on one code; your residues, plate and crystal will differ.

At a real X-ray diffractometer

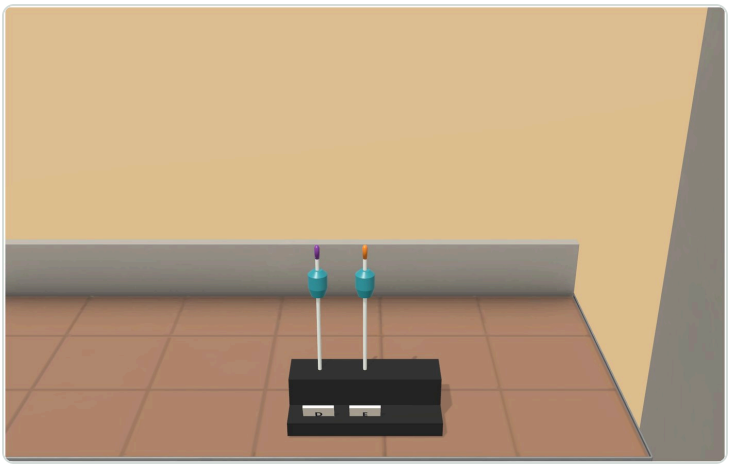
- **X-rays** are made only inside the shielded cabinet, and only with its doors shut: an interlock locks the doors while the X-rays are on, and the orange lamps show it. Never defeat an interlock.
- **Liquid nitrogen** (for freezing crystals) burns skin in seconds: face shield and cryogenic gloves, and never in a closed, small room.
- **The cold stream** blows nitrogen gas at 100 K onto the crystal: keep fingers out of it.
- The NMR hall's rules of Session 1 still hold: the 5 gauss line, nothing of steel near the magnet, the oxygen monitor.

PART A

In the NMR hall: two titrations

1 Samples D and E

The rack now holds two tubes: D (purple cap), ubiquitin with the SH3-A domain of CD2AP added step by step, and E (orange), with SH3-C. Each goes into the magnet as in Session 1: take it, climb, lower it in, lock, shim, tune, 25 °C. Then Run the **HSQC titration**: eight HSQCs, the partner added between them.



Tubes D and E in the rack.

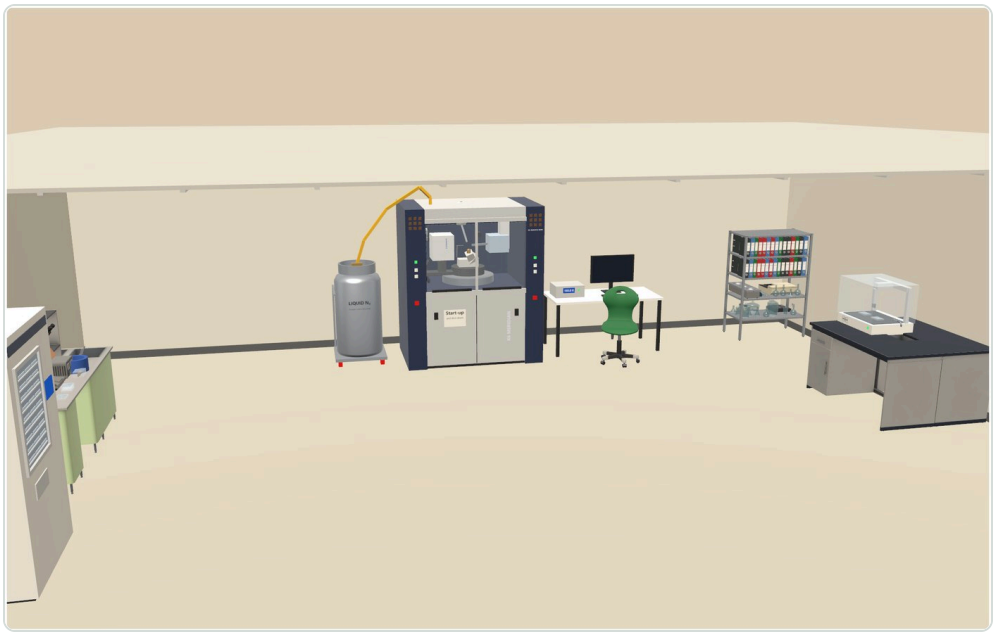
> **Why titrate?** Watching each peak move as more partner is added tells you which residues the partner touches, and how tightly it binds; it needs only the labelled protein and the partner, no crystal and no structure calculation.

PART B

In the X-ray room

2 The room

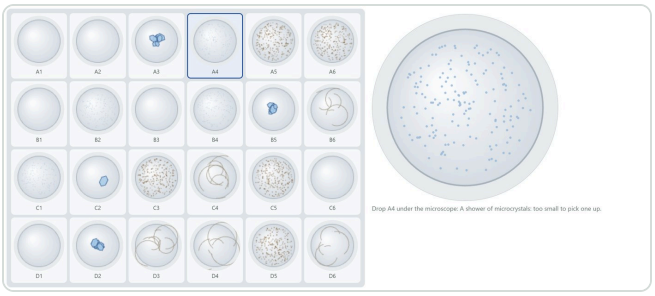
Press X-ray room. The diffractometer's cabinet stands against the back wall, with its dewar of liquid nitrogen on the left (for the cold stream) and its desk on the right. Along the left wall, the crystal bench with its microscopes, and the imaging incubator that photographs the plates for weeks; along the right, the robot that sets the drops.



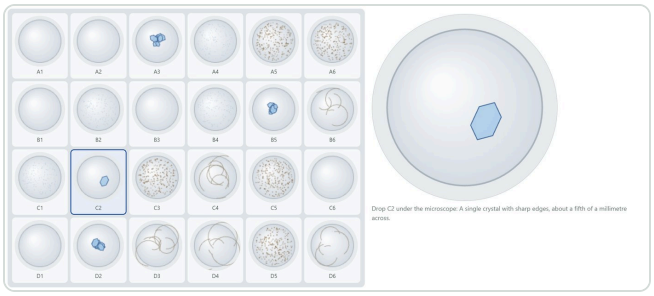
The X-ray room.

3 Choose a drop

In the page's part **Growing a crystal**, click each of the 24 drops to look at it under the microscope. Most are clear, brown precipitate, a skin, a shower of microcrystals, or crystals grown into one another. Find a drop with a single crystal with sharp edges.



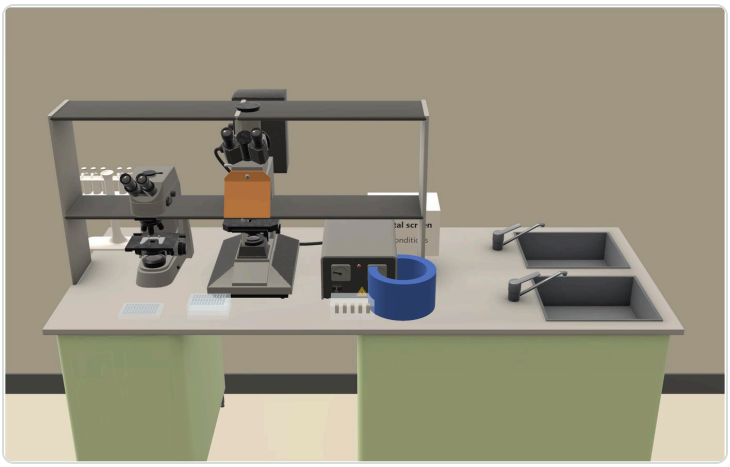
A shower of microcrystals: too small to pick one up. The room refuses to fish from it.



A single crystal: this is the one to fish.

4 Fish it, freeze it

Under **In the X-ray room**: Fish from drop … (or click the plate on the bench), then Into the nitrogen (or click the blue foam dewar).

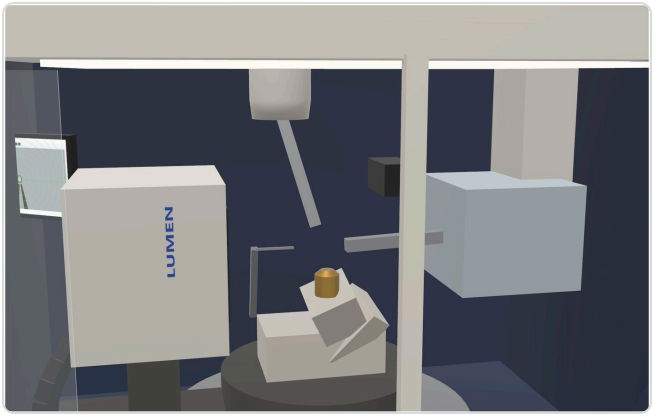


The crystal bench: the microscopes, a plate in front of them, the foam dewar of liquid nitrogen.

> **Why freeze?** At room temperature the beam's radiation damages a protein crystal within minutes; at 100 K it lasts for hours, long enough for a full data set.

5 Mount it and centre it

Open them (the doors), Mount it on the goniometer in the cold stream, Centre it in the beam. In the room you can click the doors and the goniometer instead.



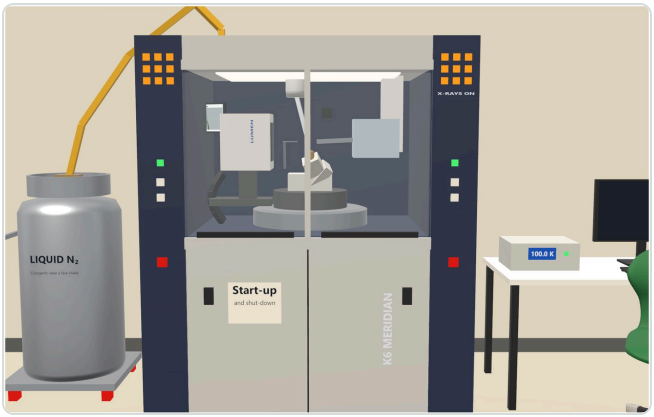
Inside the open cabinet: the detector (left), the goniometer with the crystal on its pin (middle), the cold stream from above, the X-ray source (right).

- ✓ **Fish a crystal**
  - Fish from drop C2
- ✓ **Freeze it**
  - Into the nitrogen
- ✓ **Open the doors**
  - Close them
- ✓ **Mount it on the goniometer**
  - Mount it
- ✓ **Centre it in the beam**
  - Centre it
- 6 **Close the doors, X-rays on** Next
  - The lamps on the cabinet light while the X-rays are on.
  - Close the doors
  - X-rays on
- 7 **Collect a frame**
  - Collect
  - X-rays off

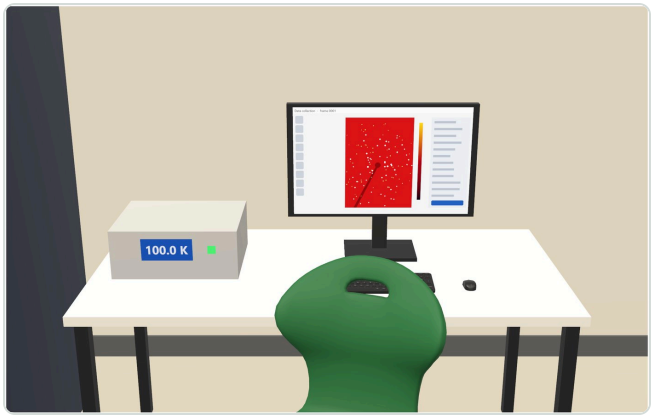
The steps, each ticked as it is done.

6 Close the doors, X-rays on, collect

Close the doors, X-rays on: the orange lamps on the cabinet light. Then Collect (or click the computer): a frame of spots appears on the desk's screen and in the part **Diffraction and resolution**. Switch the X-rays off when you are done.



The X-rays on: the lamps lit. The doors are locked.



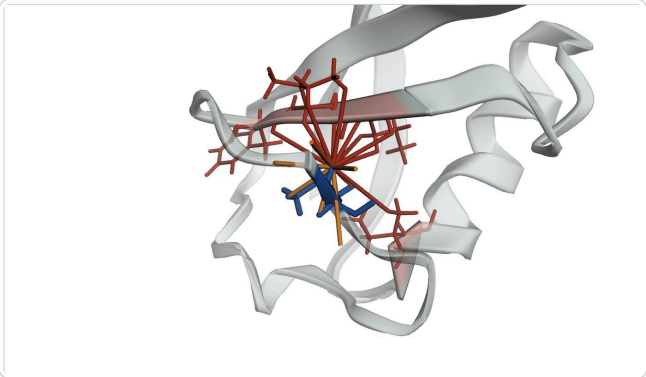
A frame on the desk's computer.

PART C

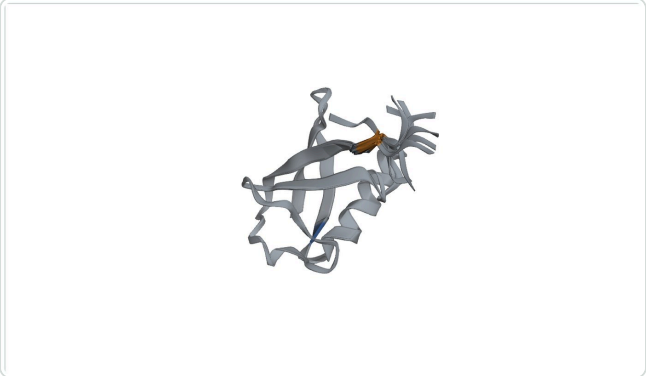
Reading what you recorded

7 From NOEs to a structure

The structure shows your residue (blue) and every proton within 5 Å of its amide proton: orange lines to residues close along the chain, red lines (and red side chains) to residues five or more away. The bundle shows the ten models; The motion ensemble, 116 models that show how the protein moves.



One residue's NOEs.

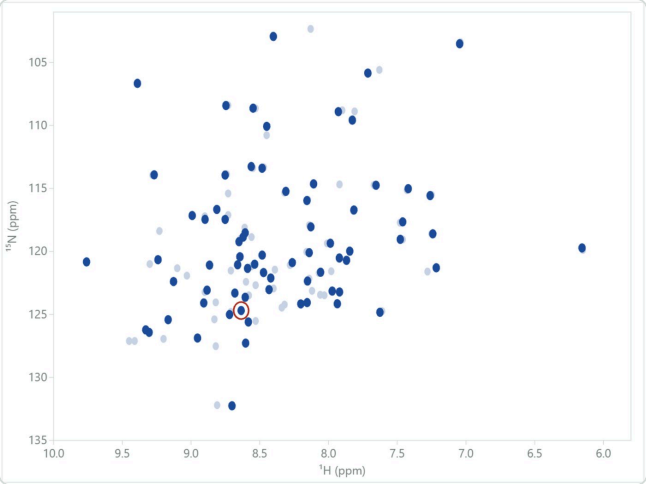


The bundle, with the two residues to compare.

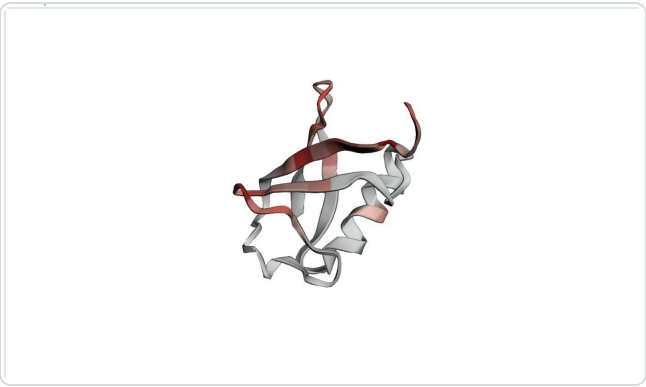
To answer: count the red lines (the long-range NOEs); turn the bundle and see where the ten models part company.

8 Binding

Choose the partner and move the slider: the peaks walk from their free places (pale) toward their bound places. Click a peak to read its shifts. The structure below is coloured by how far each residue's peak moves.



The titration part way: your residue ringed in red.

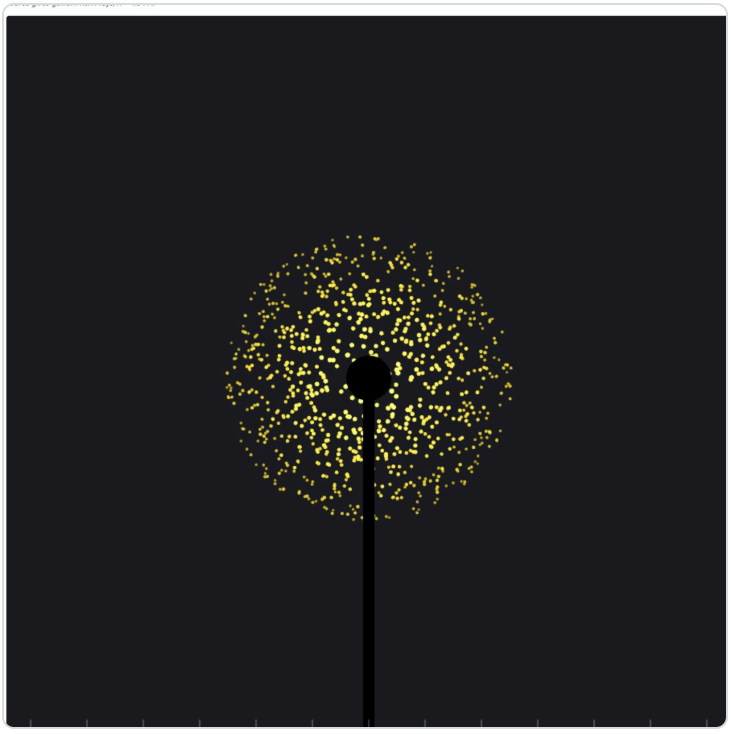


The structure coloured by the moves.

To answer: read your residue's peak with no partner (slider at 0) and at saturation (slider at 3), and use $\Delta\delta = \sqrt{(\Delta\delta_H^2 + (0.14\ \Delta\delta_N)^2)}$. Find the peak that travels furthest for each partner; compare the two coloured structures.

9 Diffraction and resolution

The frame, with a millimetre scale from the beam's centre. The page gives the detector's distance D and how far out the furthest spots lie, r.

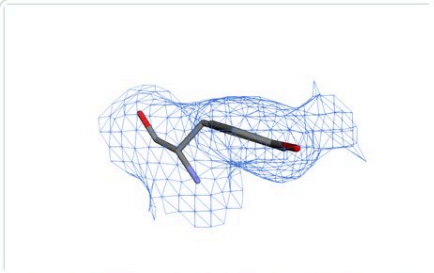


One frame (yours will differ). The shadow is the beamstop and its arm.

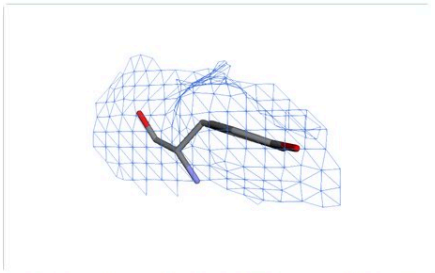
To answer: $2\theta = \arctan(r / D)$; then $d = \lambda / (2 \sin \theta)$, with $\lambda = 1.34$ Å. The prelab works one.

10 The electron density, and the B-factor

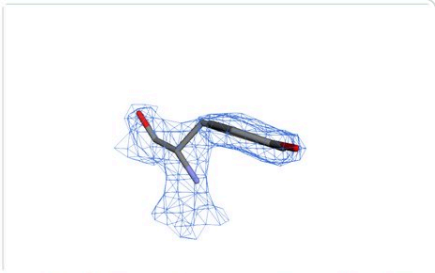
Three maps of the same ring residue at 3.5, 2.5 and 1.5 Å, in a shuffled order, then the crystal structure coloured by B-factor (blue low, red high).



Map 1: the density round Tyr59 and its neighbours, with the atoms built into it.

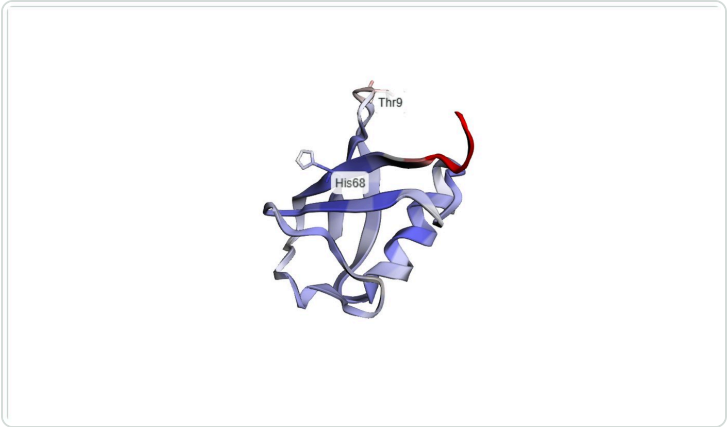


Map 2: the density round Tyr59 and its neighbours, with the atoms built into it.



Map 3: the density round Tyr59 and its neighbours, with the atoms built into it.

Three maps (yours are of another residue, in another order).

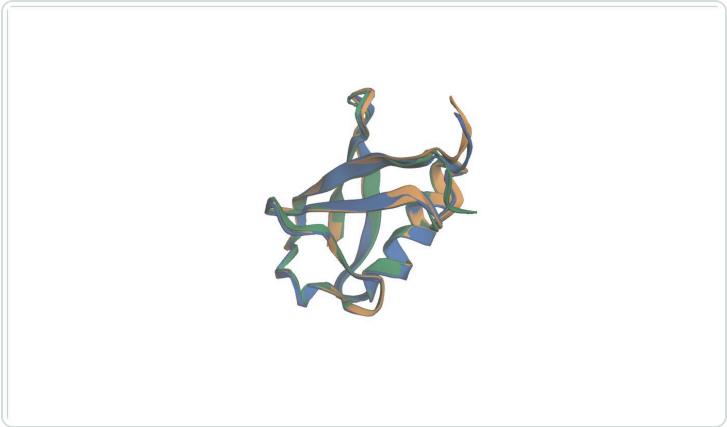


The crystal structure coloured by B-factor, the two residues labelled.

To answer: the finer the map, the tighter it follows the atoms; only at the finest does a ring show its hole. The better-ordered residue is the bluer one.

11 Crystal, solution, prediction, and where the field is now

The crystal (blue), the NMR structure (orange) and AlphaFold's prediction (green), laid on one another, with the RMSD of each pair. Then read **Where the field is now** and answer its two questions.



The three structures. Turn them to find where they part.

12 Your report

Answer the three questions in your own words, type your name, and press Print the report, or save it as a PDF. Upload the PDF where your instructor asks.

WHEN THE ROOM SAYS NO

The X-ray room's messages

The message	What to do
Clear, precipitate, a skin, microcrystals, crystals grown into one another…	Choose another drop: one with a single crystal.
You have no crystal yet: fish one from a drop.	Choose a drop in **Growing a crystal**, then fish.
Freeze the crystal first.	Into the nitrogen.
Open the cabinet's doors first.	Open them.
Mount the crystal on the goniometer first. / Mount the crystal and centre it in the beam first.	Mount it, then Centre it.
Close the doors first: the X-rays only switch on with the doors shut.	Close the doors.
The doors are locked while the X-rays are on: switch them off first.	X-rays off, then open.
Switch the X-rays on first.	X-rays on.

Quick reference

NMR hall

1. Tube D: in, set up, run the HSQC titration.
2. Tube E: the same.

X-ray room

3. Choose a drop with a single crystal.
4. Fish it; freeze it in the nitrogen.
5. Open the doors; mount; centre.
6. Close the doors; X-rays on; collect.

Reading

7. NOEs: count the long-range ones; compare the bundle.
8. Binding: a peak's move; the furthest mover for each partner.
9. Resolution: $2\theta = \arctan(r/D)$, $d = \lambda / (2 \sin \theta)$.
10. Maps: tight and holed is fine; blobs are coarse.
11. B-factor: bluer is better ordered.
12. Compare the three; the field now; the report.

The rooms are a simulation: BROKER, Pinnacle, K6 MERIDIAN and LUMEN are made-up names. The plate is made up; the binding shifts are measured (BMRB 19447, 18737); the maps are computed from the atoms of the crystal structure, PDB 1UBQ.